

Virtual Touch Modulates Perception of Pleasant Touch

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ABSTRACT

Pleasant touch by gentle stroking is processed by C-tactile afferents, and important for emotional and social touch in human communication. We aimed to investigate the effects of visual touch in a virtual environment on the perception of pleasantness of tactile touch. Five velocities were used for the tactile brushing and virtual brushing (0.3 – 30 cm/s), and participants answered the perceived pleasantness of the tactile touch irrespective of visual stimulus. We found that the perception of pleasant touch was significantly modulated by the velocity of visual brushing. Thus, the pleasant touch would be perceived by integrating vision and tactile perception.

Keywords: Pleasant touch, pseudo touch, cross modal integration.

Index Terms: I.3.7 [Computer graphics]: Three-Dimensional Graphics and Realism—Virtual Reality; H.1.2 [Models and Principles]: User/Machine Systems—human factors

1 INTRODUCTION

Interpersonal touch or social touch modulates humans' social behaviors and affects humans' emotions [1]. For example, touch enhances helping behaviors [2][3], modulates personal assessments [4], and boosts interpersonal attractiveness [5]. Prosocial behavior is also encouraged by hug from a robot [6]. Humans can discriminate and communicate several emotions such as anger, fear, disgust, love, gratitude, and sympathy [7][8].

Human C-tactile afferents respond vigorously to gentle skin stroking at intermediate velocities (1–10 cm/s), and their firing rates correlate with the ratings of subjective pleasantness [9]. Velocity of stroking a partner or baby is equivalent to the velocity that can activate C tactile afferents [10]. Thus, C-tactile afferents is a neural basis of processing gentle and pleasant tactile stimulation for social touch.

Even in virtual environments such as VRChat, people perform various types of physical contacts, and they feel as the physical contacts improve interpersonal attractiveness and communication [11]. However, the physical contacts in VRChat are not actual touches, but virtual touches. Humans can feel illusory touch as known as pseudo touch sensation by modulating speed of visual stimuli [12]. Human perceptual system integrates multi-modal sensations [13]. Thus, we hypothesized that virtual or visual

stroking on an avatar's skin modulates the perception of pleasant touch on participant's skin.

We aimed to test this hypothesis by an experiment with varying the velocities of tactile stroking and visual stroking in a virtual environment.

2 METHODS

2.1 Participants

Eighteen university students (All males, mean age 22, SD 2.1) participated in the experiment. All participants were unaware of the hypothesis of the experiments, and had normal or corrected-to-normal vision and touch sensation. They were provided informed consent before the experiment. The methods of the experiment were approved by the Ethical Committee for Human-Subject Research at the Toyohashi University of Technology.

2.2 Apparatus

We used a computer and Unity (2020) to create and present a virtual environment to a head mounted display (HMD: Valve Index, 1440 [width] × 1600 [height] pixels, 130 deg in visual angle, refresh rate 120 Hz, Figure 1). A robot (Dobot Magician Lite, position accuracy 0.2 mm) was used to control movements of a brush. To prevent the sounds of the robot from hearing a white noise (61 dBA) was presented to them during the experiment by a noise-cancelling headphone (SONY WI-SP600N).



Figure 1: Apparatus.

2.3 Stimuli and Conditions

We set five levels of velocity of tactile brushing and visual brushing

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(0.3, 1.0, 3.0, 10.0, 30.0 cm/s). Thus, all combinations of velocities were 25 in total. A real brush and a virtual brush moved at 10 cm on the left forearm of the participant (see Figure 1) and an avatar's left forearm (Figure 2). The real brush and the virtual brush started to move at the same time, and one at faster velocity stayed at the end point until the other at slower velocity stopped if the velocities were different between tactile and visual stimuli.

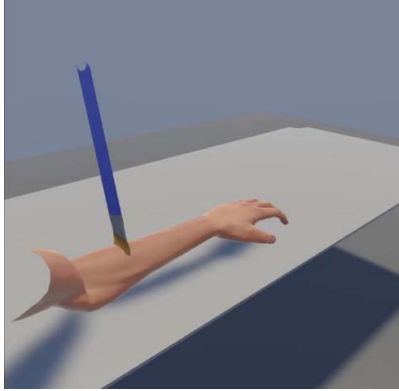


Figure 2: Visual stimulus on the HMD.

2.4 Procedure

Before the experiment, we adjusted the force of the tactile brush to the forearm at 0.4 N with a pressure sensor for each participant. We instructed participants to look at the left forearm during trials, then asked them to answer how pleasant the tactile touch was with a visual analog scale (VAS), irrespective of the visual stimulation. Each participant performed three repetitions of all 25 conditions (5 tactile x 5 visual velocity conditions) in random order. The VAS data were digitized into -100 – +100 after the experiment.

3 RESULTS

Two-way repeated measures ANOVA with Greenhouse-Geisser correction was conducted for the pleasantness measured by VAS. We found a significant main effect of the tactile-velocity condition ($F(2.06,35.05)=19.14$, $p<0.0001$, $\eta_p^2=0.530$) and the interaction of tactile and visual conditions ($F(3.13,53.17)=13.42$, $p<0.0001$, $\eta_p^2=0.441$, Figure 3). Thus, the perception of pleasant touch was modulated by the velocity of visual brushing.

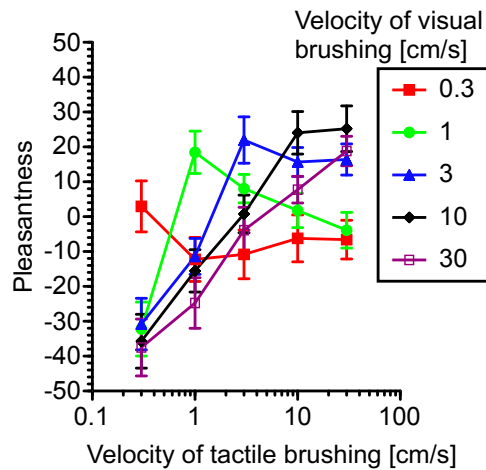


Figure 3: Results of experiment.

4 DISCUSSION

The results showed that the velocity of visual brushing on the avatar's arm in the virtual environment affected subjective ratings of pleasant touch even though the participants were asked to answer tactile pleasantness irrespective of visual stimulation. The pleasantness was basically high when the tactile and visual velocities were identical, and those scores showed inverted-U shape similarly to the C-tactile neural responses. Thus, the pleasantness measure in the experiment would be related to C-tactile processing. It is suggested that the pleasant touch would be perceived by integrating vision and tactile perception.

Our findings contribute to investigate a method to provide emotional touch by combining visual and tactile stimulations in virtual reality in addition to understand neurocognitive mechanisms of visual-tactile integration in emotional touch.

ACKNOWLEDGMENTS

This research was partly supported by JST ERATO JPMJER1701 (Inami JIZAI Body Project), and JST OPERA JPMJOP1834.

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